# BANK MARKETING PREDICTION

CIS 9660 - Data Mining for Business Analytics

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# 1. Introduction

This project utilizes the bank marketing dataset from the UCI Machine Learning Repository. The data relates to direct marketing campaigns (telephone calls) made by a Portuguese banking institution. The goal of this project is to predict whether the bank clients will subscribe to a term deposit during the campaign or not, which serves as the target variable. The subscription rate of the whole dataset was 11.27%, which indicates the imbalanced nature of the dataset. The features consist of the client's characteristics, including age, occupation, marital status, education level, etc., and information about communication with each client during this and the previous campaign, such as types of contact, contact month and weekday, number of contacts, etc. Additionally, it includes social and economic indicators during the campaign period, such as the consumer price index, 3-month Euribor rate, etc. For more information regarding the features, please refer to this project's notebook.

# 2. Exploratory Data Analysis

Before implementing any machine learning model, it's important to perform Exploratory Data Analysis (EDA) to get a better understanding of the data. In this section, I utilized different visualization techniques, including histogram, boxplot, count plot, point plot, line plot, and correlation matrix to extract insights and patterns from the data.

## 2.1. The Influence of Clients characteristics on Subscription

Below are the summaries of the insights extracted from EDA:

- Most of the clients were middle-aged, and clients aged below 30 or above 50 were more inclined to subscribe.
- The clients had various occupations, education levels, and marital statuses.
- In terms of clients' occupations, the three largest occupations among those contacted during this campaign were admin, blue-collar, and technician, but they had relatively lower subscription rates. In contrast, though the number of retired and student clients was lower, they had relatively higher subscription rates. In particular, although the blue-collar occupation was the second-largest group contacted during the campaign, it had the lowest subscription rate. Conversely, students, who were the second-smallest group, had the highest subscription rate.
- In terms of clients' marital status, although most of them were married, this group had the lowest subscription rate among those contacted. Individuals with a single marital status or unknown status had relatively higher subscription rates.
- In terms of clients' education, illiterate individuals had the highest subscription rate, but they had the smallest client base. Individuals with university degrees represented the largest client base, and the subscription rate ranked third among different education levels.
- 50% of the clients had housing loans, while the majority did not have personal loans. Clients with housing loans and personal loans had relatively higher subscription rates.
- Most of the clients did not have any defaulted credits, and those clients were more likely to subscribe, whereas individuals with defaulted credits did not subscribe at all.

With these insights, the next time when the bank conducts marketing, the bank can allocate more resources to clients with characteristics that have a positive influence on the subscription and tailor the marketing content to these clients, improving the success rates.

### 2.2. The Influence of Contact Ways and Time on Subscription

Below are the summaries of the insights extracted from EDA:

- Clients contacted via cellular phone were more likely to subscribe than those contacted by telephone. Most of the phone calls lasted less than 5 minutes, and clients who subscribed tended to have longer phone call durations.
- On average, clients were contacted 2.5 times during the campaign, and the majority of them were not contacted during the last campaign. Subscribed clients had fewer contacts during this campaign, but more contacts during the last campaign. Clients who subscribed successfully in the previous campaign were more likely to subscribe to this one as well.
- The campaign was active for 10 months from March to December, and more than 25% of the clients were contacted in May. Most clients were contacted between May and August, but subscription rates were relatively lower during this period. The weighted average subscription rates were higher in March, September, October, and December when the employment variation rate, consumer price index, consumer confidence index, Euribor 3-month rate, and the number of employees were lower.
- Euribor 3-month rate, employment variation rate, and number of employees are highly correlated and Euribor 3-month rate, employment variation rate, and consumer price index are moderately correlated.
- Client subscription rates were lowest on Monday and highest on Thursday.

With these insights, the bank can choose the right timing to conduct the market campaign and choose the right ways to communicate with the clients to achieve higher subscription rates, hence improving the efficiency of marketing campaigns and reducing costs.

## 3. Machine Learning

Although we can draw insightful conclusions from exploratory data analysis, machine learning can be utilized to predict the probability of subscription of each client, which helps the bank to quantify the success rate of each client and allocate the resources even more efficiently. In this section, I implemented various machine-learning techniques to find the optimal model for predicting client subscriptions, including data preprocessing, cross-validation, and model selection.

### 3.1. Data Preprocessing

As the dataset includes some categorical features, one-hot encoding, which creates a new binary column for each category and encodes the categories as 0 or 1, is required to prepare the data for the model. In addition, the numerical features in the dataset have very different scales. This requires the use of a standardization scaler which standardizes the features of a dataset so that they have the properties of a standard normal distribution, with a mean of 0 and a standard deviation of 1. This is very helpful for some linear algorithms, such as logistic regression and K-Nearest Neighbors.

After encoding and standardization, the data set was split into the training set and test set, with a test size of 30% and stratified sampling based on the distribution of the target variable as the dataset was very imbalanced.

### 3.2. Cross-Validation and Model Selection

This project used an ensemble of machine learning algorithms as below:

- Logistic Regression: Logistic Regression estimates the probabilities using a logistic function, ensuring the output lies between 0 and 1. The labels are assigned based on whether the probability is higher or lower than 0.5.
- Naive Bayes: A probabilistic algorithm based on Bayes Theorem, with the naive assumption of independence between every pair of features. It is favored for its computational efficiency, but the violation of the assumption will result in poor model performance.
- Decision Trees: Decision Trees split the dataset into subsets based on Gini Impurity or Entropy. They are easily visualized and understood by people with no technical background but can suffer from overfitting if not properly tuned.
- Random Forests: A Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the mode of the classes or mean prediction of the individual trees, so it's robust against overfitting.
- Support Vector Machines (SVM): SVM finds the hyperplane that best divides a dataset into classes. It's effective in high-dimensional spaces and is known for its effectiveness in classification, especially in binary problems.
- K-Nearest Neighbors (KNN): The output is classified based on the majority of the k-nearest points in the feature space. It is computationally expensive with large datasets.

These algorithms are commonly used in classification problems and have different characteristics. To find the optimal algorithms for making predictions on this bank marketing dataset, I performed cross-validation with 5 folds and used accuracy as the scoring metric. The results showed that Naive Bayes had the lowest average accuracy of 0.5722, and Decision Trees had an average accuracy of 0.8386. Given the subscription rate was 11.27%, any model with an accuracy lower than 0.8873 is not useful for predicting. Therefore, these two algorithms were excluded after cross-validation.

### 3.3. Model Training and Validation

Since the remaining models had similar cross-validation average accuracies and were all greater than 0.8873, I trained them with the training set and tested them on the unseen test set, and generated classification reports to find out which model performed best on this bank marketing dataset. It turned out that the test accuracies of each model were very close, and didn’t deviate from the cross-validation scores, indicating the models didn’t overfit or underfit.

To choose the best model for future production use, as our goal is to help the bank to increase the subscription rate, we want the model to have the highest precision on the truly subscribed clients. According to the results, the Support Vector Machines model achieved the highest precision of 0.68 on the truly subscribed clients, which means that 68% of the clients that the model predicted as subscribed were truly subscribed clients.

## 4. Conclusion

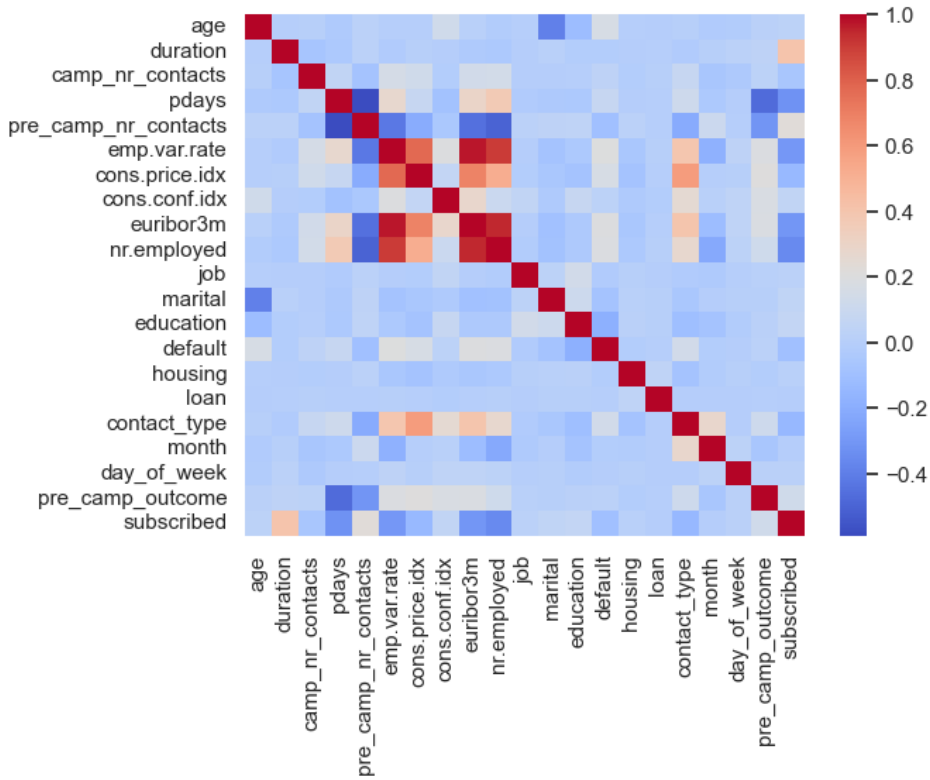
By utilizing the insights from the EDA, and with the help of the SVC model, the bank would be able to better target clients who are more likely to subscribe to term deposits based on their characteristics and optimize the timing and ways of marketing.

However, we should note that although SVC achieved the best result, it’s computationally expensive. If we prefer speed over accuracy, Logistic Regression could be an alternative solution since its performance is very close to SVC and is more efficient when dealing with a larger dataset in the future.

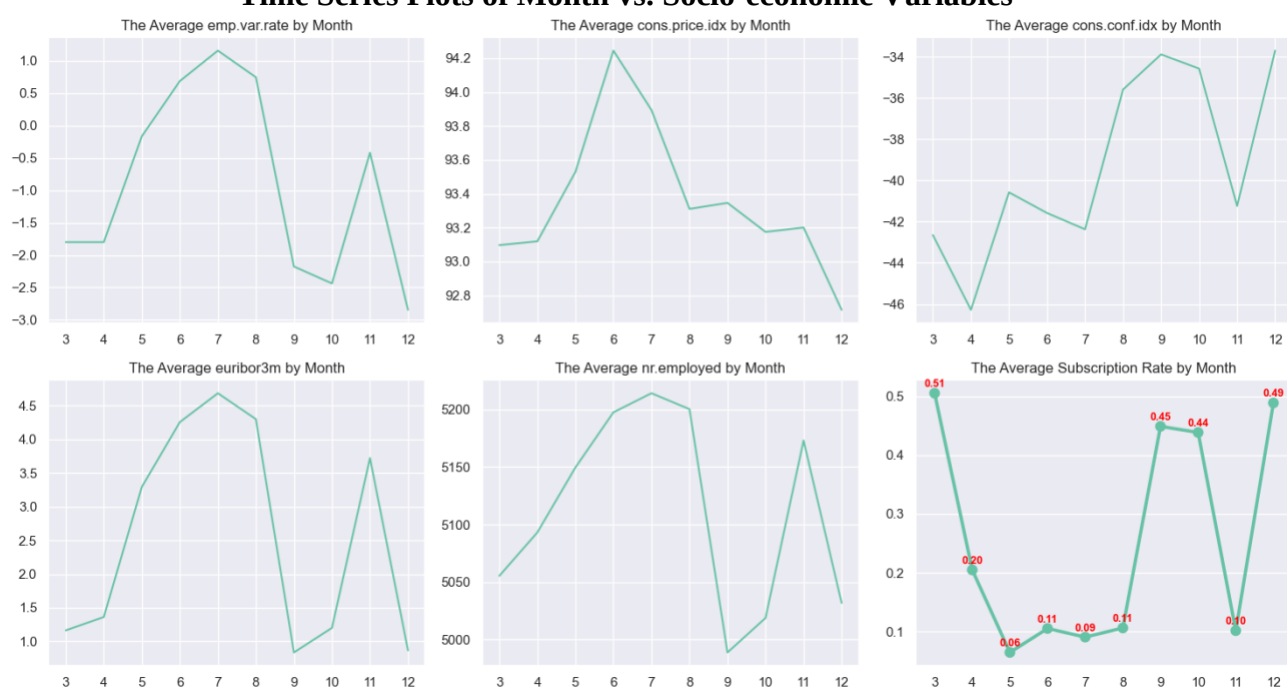
## Appendix

```
Classification Report of LogisticRegression:
              precision    recall  f1-score   support

           0       0.91      0.99      0.95     10965
           1       0.67      0.23      0.34      1392

    accuracy                           0.90     12357
   macro avg       0.79      0.61      0.64     12357
weighted avg       0.88      0.90      0.88     12357

------------------------------------------------------------
Classification Report of KNeighborsClassifier:
              precision    recall  f1-score   support

           0       0.91      0.97      0.94     10965
           1       0.52      0.25      0.34      1392

    accuracy                           0.89     12357
   macro avg       0.72      0.61      0.64     12357
weighted avg       0.87      0.89      0.87     12357

------------------------------------------------------------
Classification Report of RandomForestClassifier:
              precision    recall  f1-score   support

           0       0.92      0.97      0.94     10965
           1       0.56      0.31      0.40      1392

    accuracy                           0.89     12357
   macro avg       0.74      0.64      0.67     12357
weighted avg       0.88      0.89      0.88     12357

------------------------------------------------------------
Classification Report of SVC:
              precision    recall  f1-score   support

           0       0.91      0.99      0.95     10965
           1       0.68      0.21      0.32      1392

    accuracy                           0.90     12357
   macro avg       0.80      0.60      0.63     12357
weighted avg       0.88      0.90      0.88     12357
```

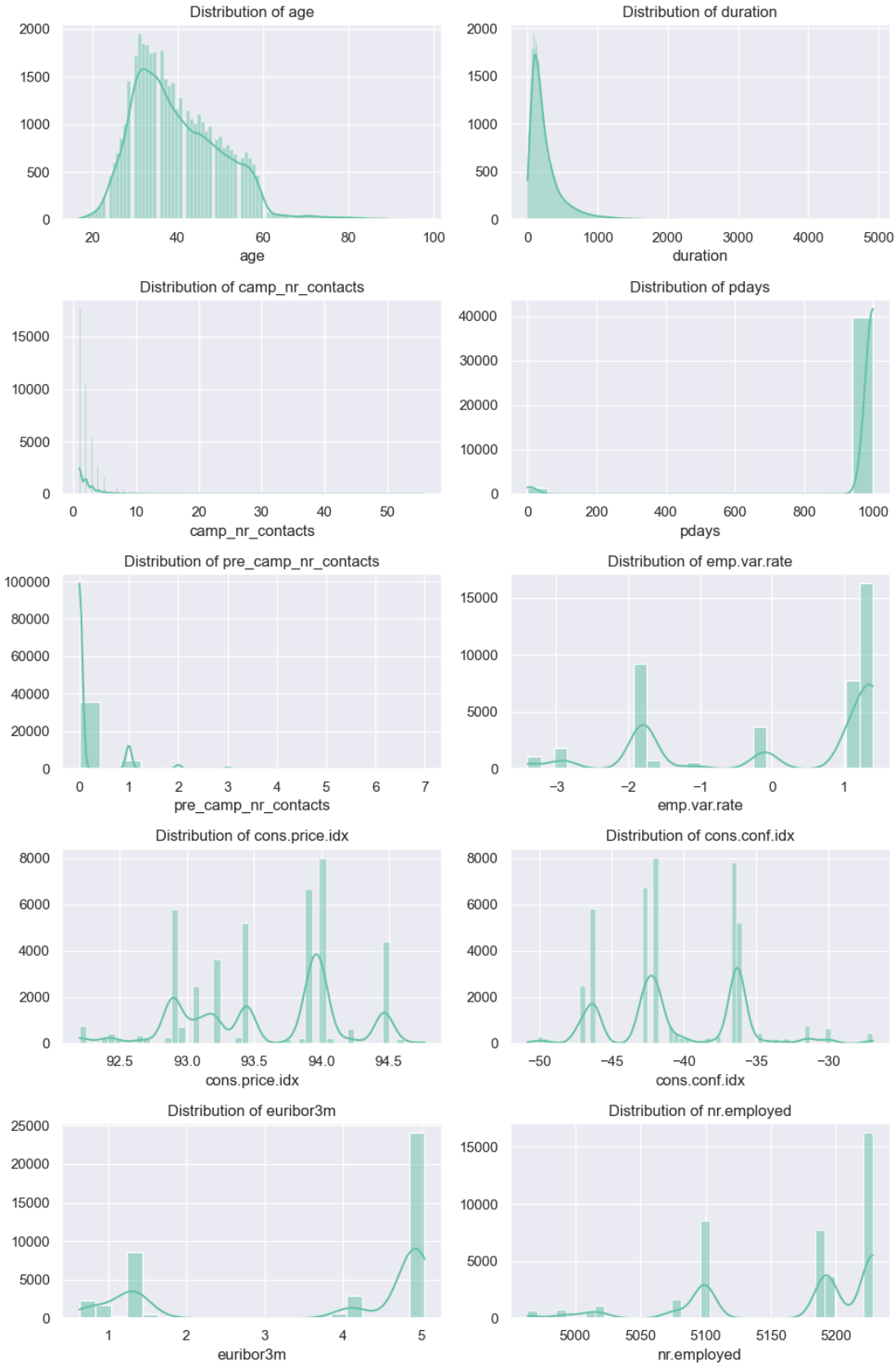
## Correlation Matrix



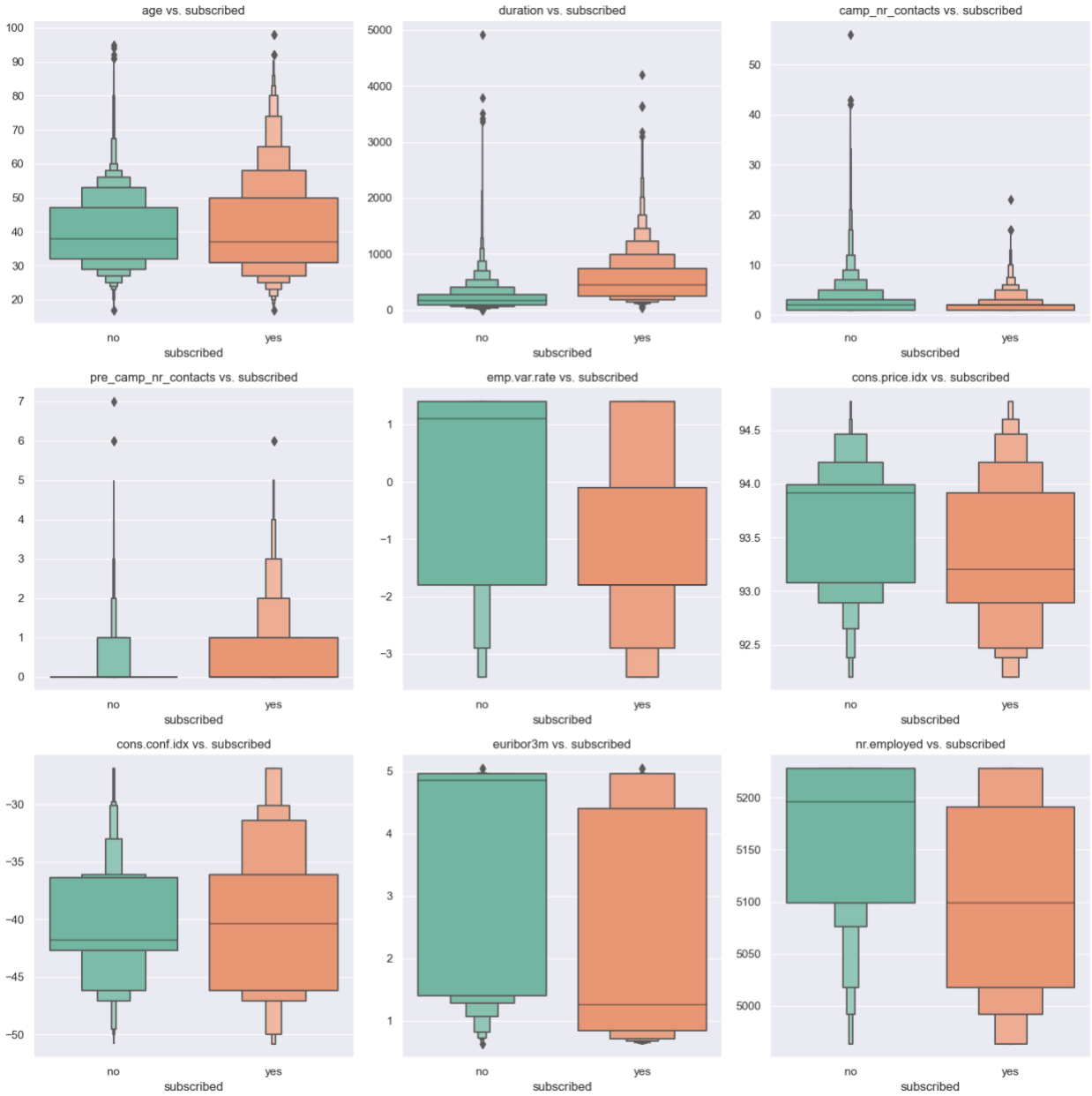

## Time Series Plots of Month vs. Socio-economic Variables



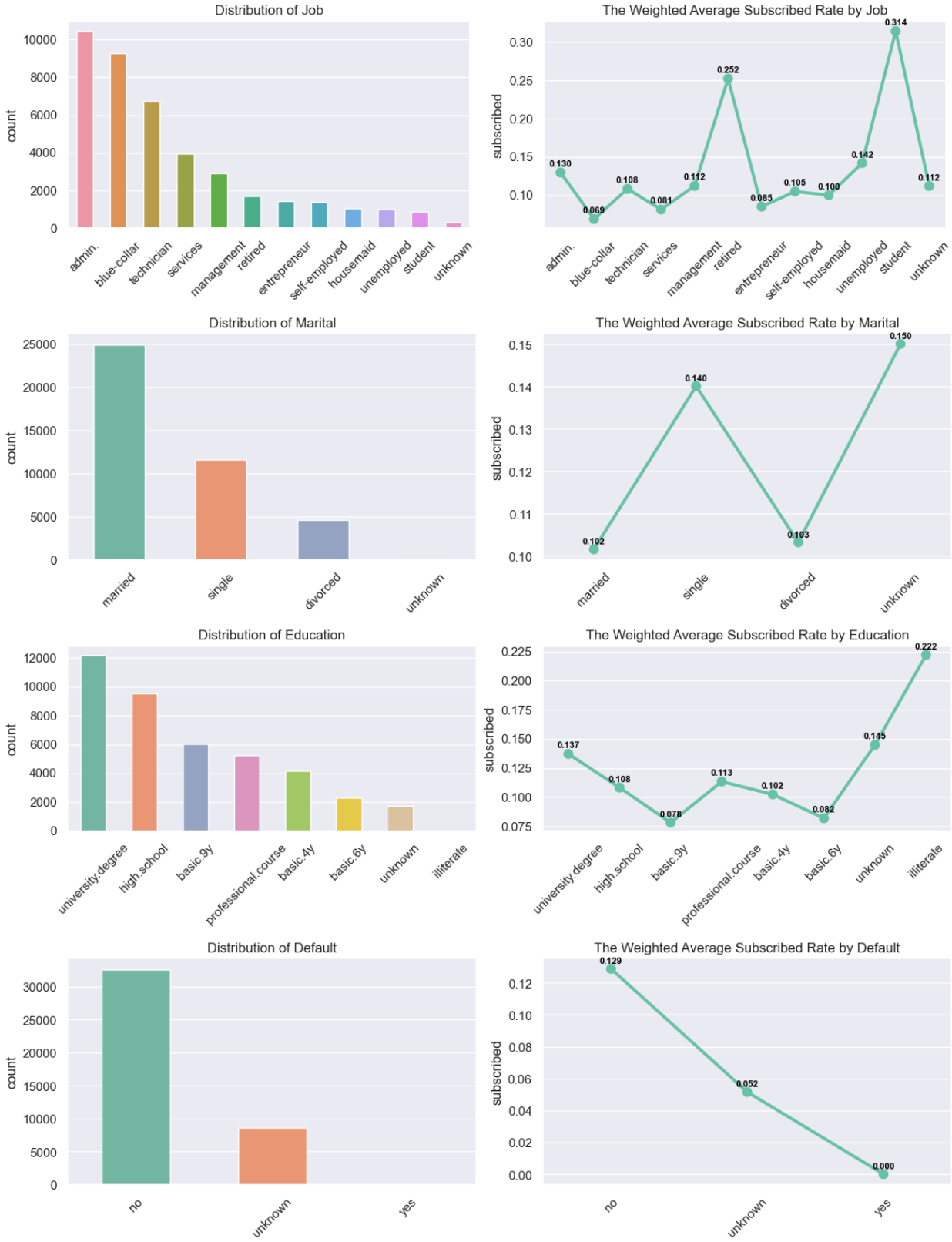
## Distribution of Numerical Features



## Boxenplots of Numerical Variables vs. Target Variable



## Count Plots and Point Plot of the Subscription by Categorical Variables

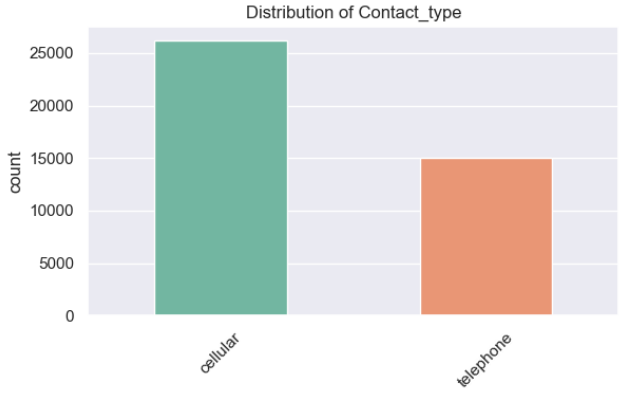


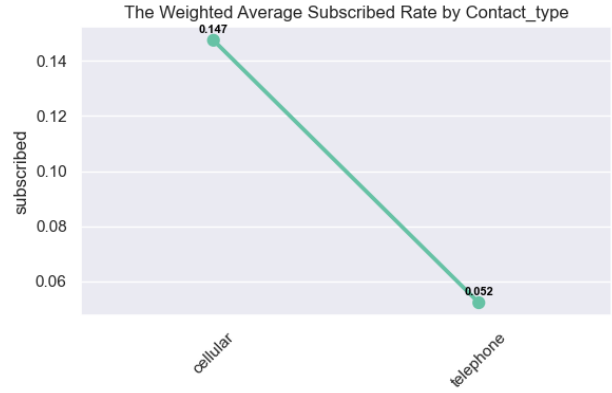
Distribution of Housing
count
20000
15000
10000
5000
0
yes
no
unknown

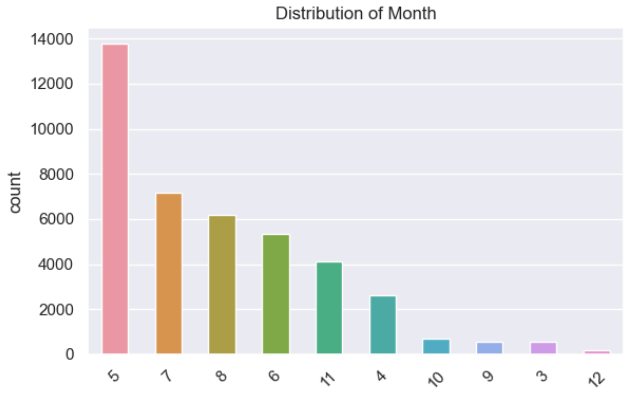
The Weighted Average Subscribed Rate by Housing
subscribed
0.116
0.115
0.114
0.113
0.112
0.111
0.110
0.109
0.108
0.116
0.109
0.108
yes
no
unknown

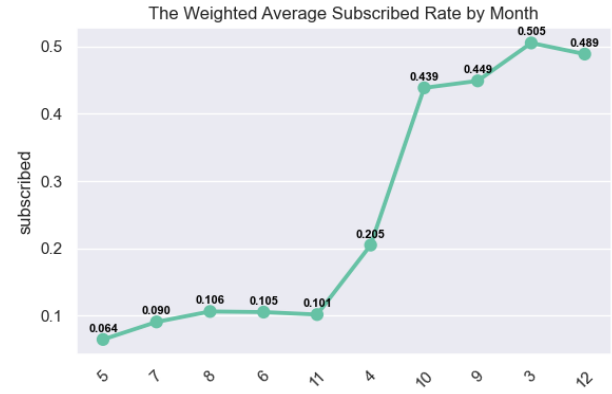
Distribution of Loan
count
35000
30000
25000
20000
15000
10000
5000
0
no
yes
unknown

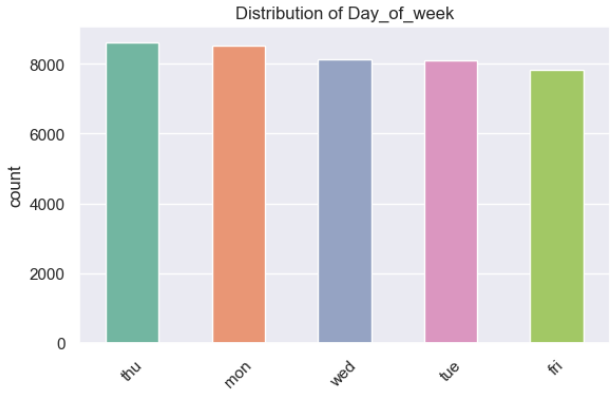
The Weighted Average Subscribed Rate by Loan
subscribed
0.113
0.112
0.111
0.110
0.109
0.108
0.113
0.109
0.108
no
yes
unknown

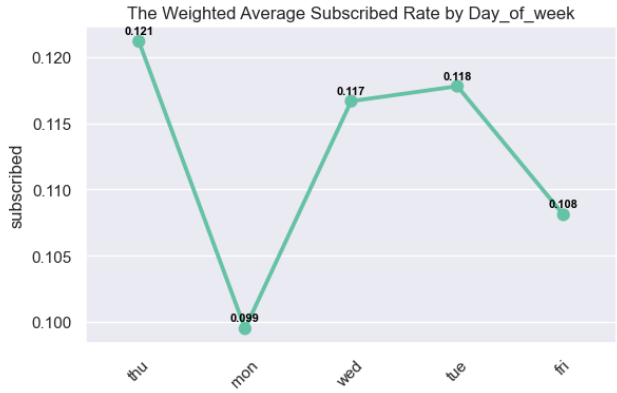

Distribution of Contact_type
count
25000
20000
15000
10000
5000
0
cellular
telephone

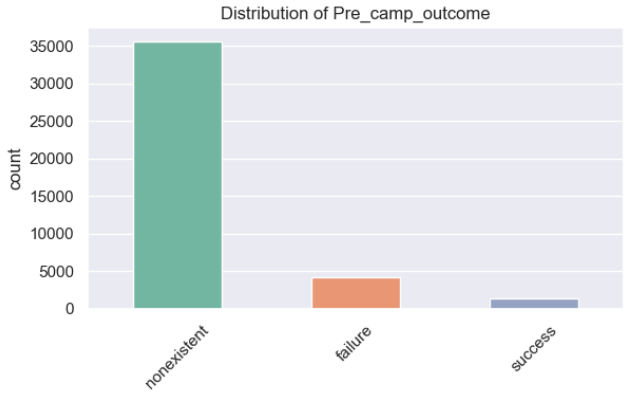

The Weighted Average Subscribed Rate by Contact_type
subscribed
0.147
0.14
0.12
0.10
0.08
0.06
0.147
0.052
cellular
telephone

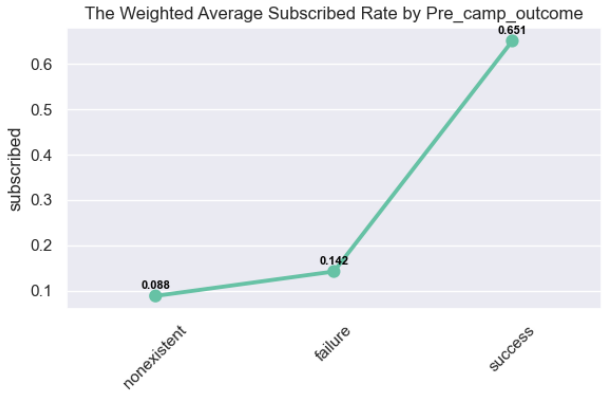

Distribution of Month
count
14000
12000
10000
8000
6000
4000
2000
0
5
7
8
6
11
4
10
9
3
12

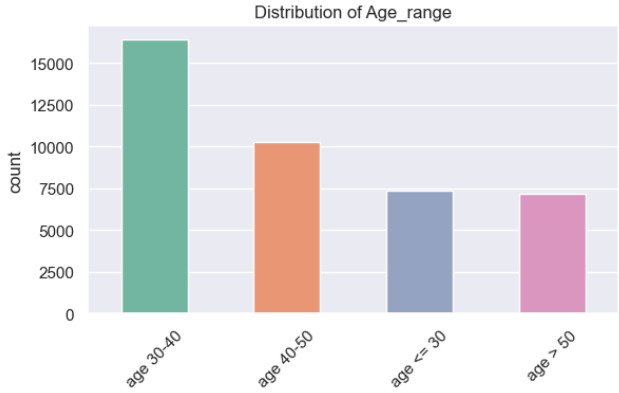

The Weighted Average Subscribed Rate by Month
subscribed
0.5
0.4
0.3
0.2
0.1
0.064
0.090
0.106
0.105
0.101
0.205
0.439
0.449
0.505
0.489
5
7
8
6
11
4
10
9
3
12


Distribution of Day_of_week
count
0
2000
4000
6000
8000
thu
mon
wed
tue
fri


The Weighted Average Subscribed Rate by Day_of_week
subscribed
0.100
0.105
0.110
0.115
0.120
0.121
0.099
0.117
0.118
0.108
thu
mon
wed
tue
fri

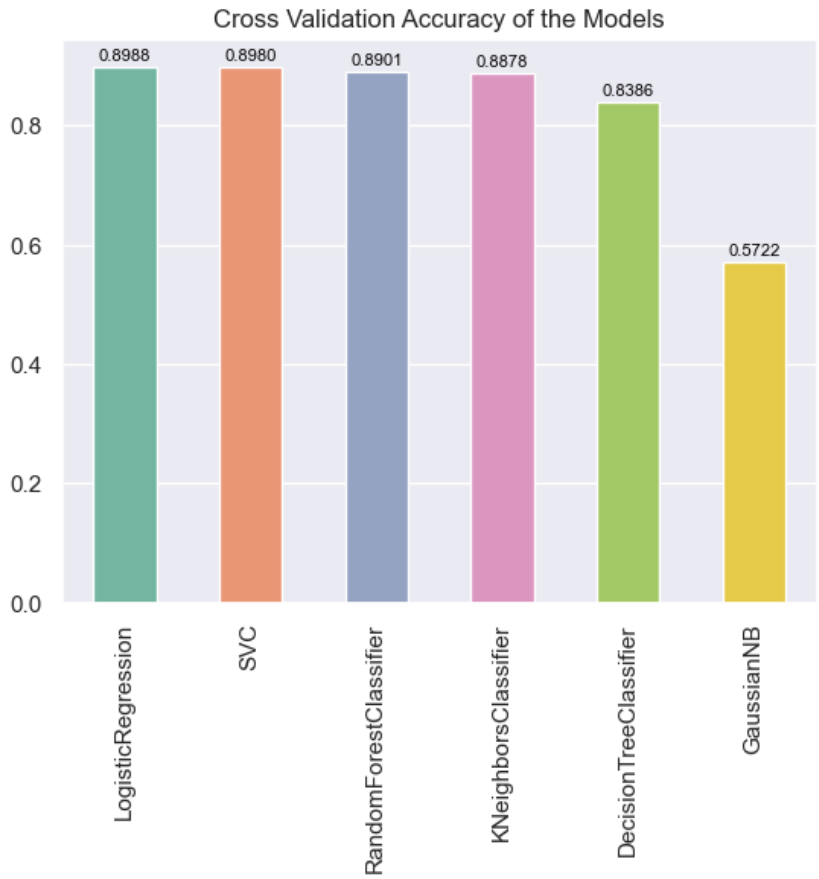

Distribution of Pre_camp_outcome
count
0
5000
10000
15000
20000
25000
30000
35000
nonexistent
failure
success

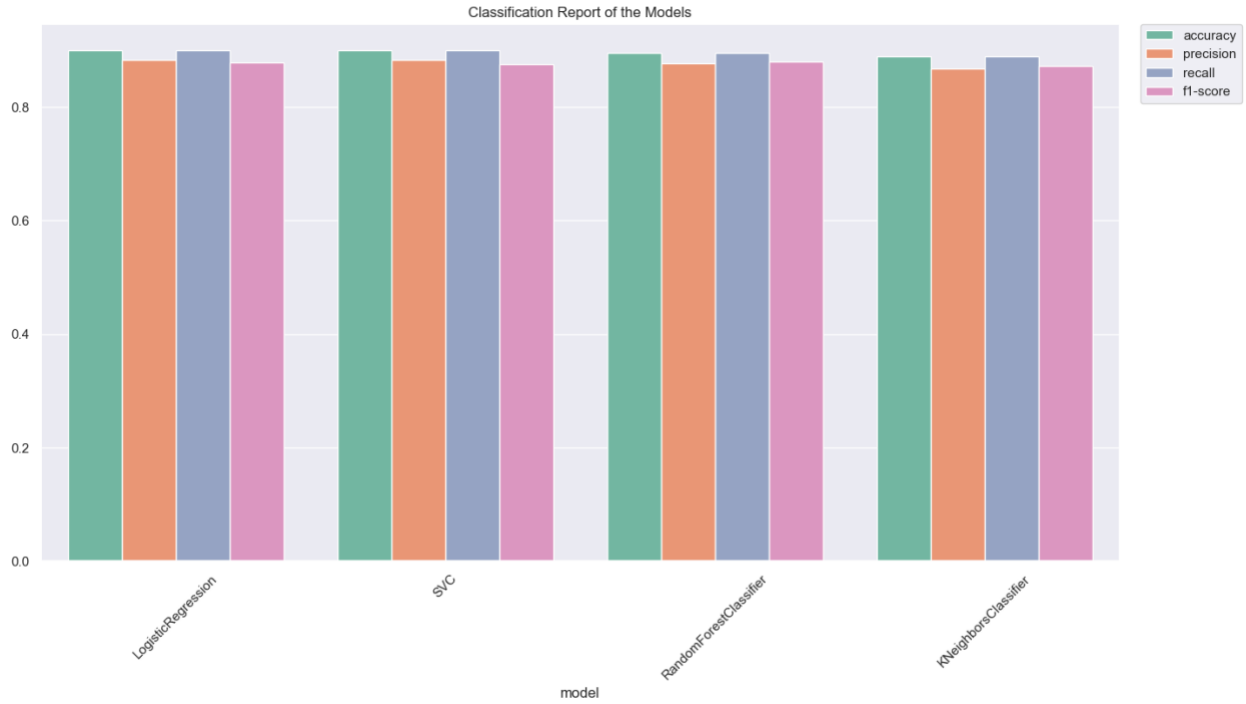

The Weighted Average Subscribed Rate by Pre_camp_outcome
subscribed
0.1
0.2
0.3
0.4
0.5
0.6
0.088
0.142
0.651
nonexistent
failure
success


Distribution of Age_range
count
0
2500
5000
7500
10000
12500
15000
age 30-40
age 40-50
age <= 30
age > 50

The Weighted Average Subscribed Rate by Age_range
subscribed
0.08
0.09
0.10
0.11
0.12
0.13
0.14
0.15
0.097
0.082
0.152
0.151
age 30-40
age 40-50
age <= 30
age > 50

## Cross-Validation Accuracy




## Classification Report